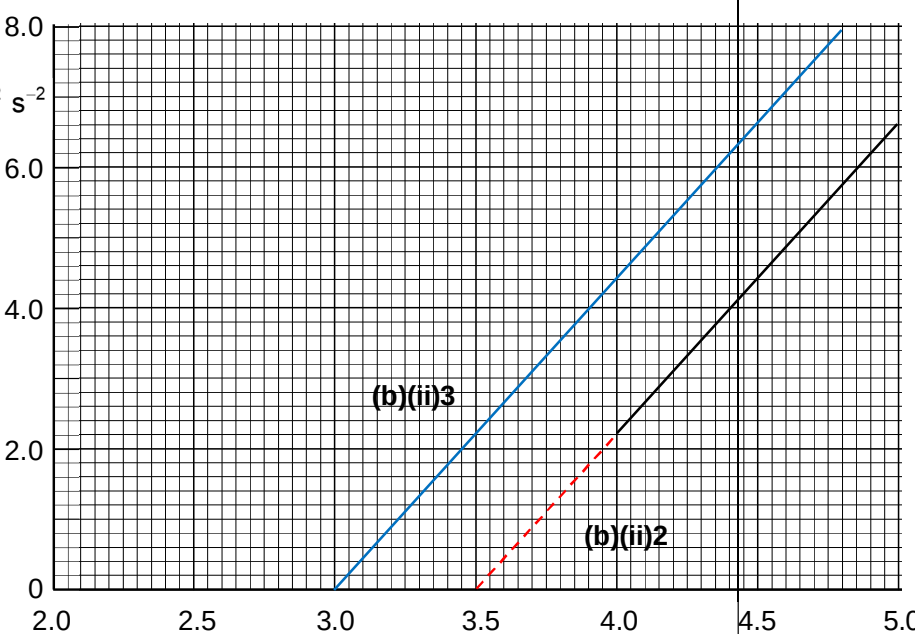


	$v_o = 1.08 \times 10^{-6} \text{ m s}^{-1}$	answer
9(a)	The work function energy of a surface is the minimum amount of energy needed to liberate an electron from its surface.	[1]
9(b)(i)1	Photoelectrons are emitted from plate A with a <u>range of kinetic energies up to a maximum value.</u>	[1]
	As the <u>potential of plate B becomes lower and lower</u> compared to plate A, <u>fewer photoelectrons can reach plate B.</u>	[1]
9(b)(i)2	$V_s = 1.0 \text{ V}$ $KE_{\text{max}} = eV_s$ $\frac{1}{2} m_e v_{\text{max}}^2 = eV_s$ $\frac{1}{2} (9.11 \times 10^{-31}) v_{\text{max}}^2 = (1.60 \times 10^{-19})(1.0)$ $v_{\text{max}} = 5.9267 \times 10^5 \text{ m s}^{-1}$ $v_{\text{max}} = 5.9 \times 10^5 \text{ m s}^{-1}$	[1] for V_s [1] for correct equation $\frac{1}{2} m_e v_{\text{max}}^2 = eV_s$ [1] for correct answer
9(b)(i)3	photoelectric current = $3.2 \times 10^{-9} = \frac{N_e}{t} (1.60 \times 10^{-19})$ $\frac{N_e}{t} = \frac{3.2 \times 10^{-9}}{1.60 \times 10^{-19}} = 2.0 \times 10^{10} \text{ s}^{-1}$	[1] for correct substitution [1] for correct answer
9(b)(i)4	$\frac{N_p}{t} = (1.0 \times 10^5) (2.0 \times 10^{10}) = 2.0 \times 10^{15} \text{ s}^{-1}$	[1] for correct substitution and correct answer

9(b)(i)5	photoelectric current / nA electric potential of plate B / V	[1] for half the current for all values of p.d. [1] for same V_s
9(b)(ii) 1	$KE_{\max} = hf - \Phi$ $\frac{1}{2} m_e v_{\max}^2 = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$ $v_{\max}^2 = \frac{2hc}{m_e} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$ $\text{gradient} = \frac{2hc}{m_e}$ $\text{From graph, gradient} = \frac{(6.6 - 2.2) \times 10^{11}}{(5.0 - 4.0) \times 10^6} = 4.4 \times 10^5$ $\frac{2hc}{m_e} = 4.4 \times 10^5$ $\frac{2h(3.00 \times 10^8)}{9.11 \times 10^{-31}} = 4.4 \times 10^5$ $h = 6.6807 \times 10^{-34} \text{ J s}$ $h \approx 6.7 \times 10^{-34} \text{ J s}$	[1] for gradient = $\frac{2hc}{m_e}$ [1] for correct calculation of gradient [1] for correct answer

9(b)(ii) 2 and 3	 From graph, $\frac{1}{\lambda_0} = 3.5 \times 10^6 \text{ m}^{-1}$ $\Phi = \frac{hc}{\lambda_0} = (6.7 \times 10^{-34})(3.00 \times 10^8)(3.5 \times 10^6)$ $\Phi = 7.035 \times 10^{-19} \text{ J}$ $\Phi = \frac{7.035 \times 10^{-19}}{1.60 \times 10^{-19}} = 4.3969 \text{ eV}$ $\Phi \approx 4.4 \text{ eV}$	$\frac{1}{\lambda} / 10^6 \text{ m}^{-1}$ [1] for correct substitution [1] for correct answer [1] for correct graph of metal with lower Φ
9(c)(i)	$\text{Power} = \frac{N_p}{t} \frac{hc}{\lambda}$ $0.76 \times 10^{-3} = \frac{N_p}{t} \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{200 \times 10^{-9}}$ $\frac{N_p}{t} \approx 7.6 \times 10^{14} \text{ s}^{-1}$	[1] for correct substitution
9(c)(ii)	Magnitude of force by metal surface on photons $= \frac{N_p}{t} \Delta p$ $= \frac{N_p}{t} \frac{h}{\lambda}$ $= (7.6 \times 10^{14}) \frac{6.63 \times 10^{-34}}{200 \times 10^{-9}}$ $= 2.5194 \times 10^{-12} \text{ N}$ $\approx 2.5 \times 10^{-12} \text{ N}$	[1] for correct substitution

	Magnitude of force by photons on metal surface $= 2.5 \times 10^{-12} \text{ N}$	[1] for correct answer
--	---	------------------------